

High Performance Scientific Computing (CD61002)

Assignment-1

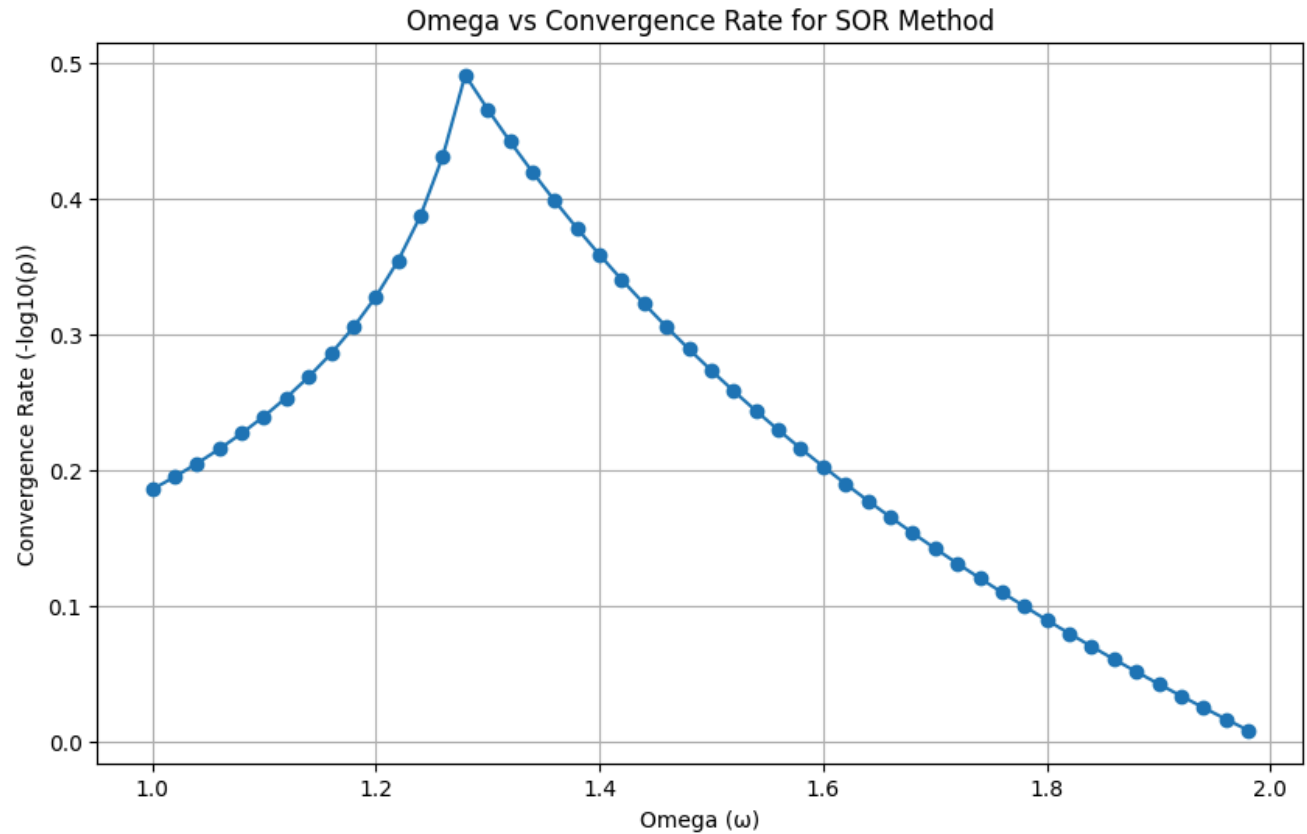
Name: Utsab Karan Roll: 23EC30058

Procedure:

1. First create a mesh on GMSH with the desired number of points. Export the mesh as a .msh file to your directory with the mshtofem.c and fem.c files.
2. Modify the boundary conditions in the fem.c file to account for the given conditions.
3. Compile both the mshtofem.c and fem.c files.
4. Run the compiled files. It generates the Kmat.txt and Fvec.txt files which contain the A and B matrices respectively.
5. Based on this I designed the iterative functions to get a solution for the equation $Ax = b$.
6. Python was used to calculate the G and f matrices and also to calculate the maximum eigenvalues. This step can also be done with Matlab or Octave. The spectral radius and convergence rates were calculated from the eigenvalues.
7. Using the generated G and f matrices the iterative methods were applied with the following results.
8. For the SOR method, the variation of convergence rate with the relaxation factor ω was also plotted.

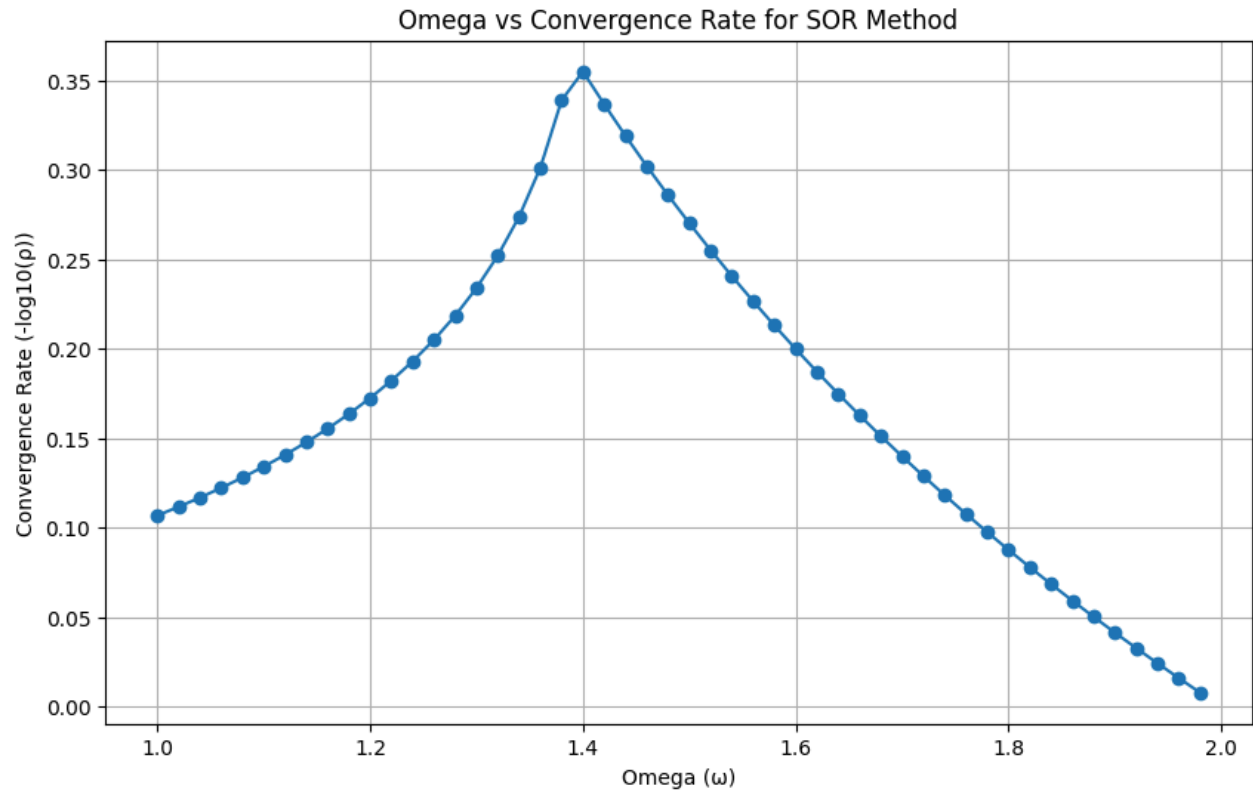
45 points on the mesh:

	Jacobi	Gauss-Seidel	GS-SOR			
			$\omega=1.1$	$\omega=1.3$	$\omega=1.6$	$\omega=1.9$
Iterations till 1.0E-06	57	30	24	14	34	174
Spectral Radius	0.804	0.652	0.576	0.342	0.627	0.907
Convergence Rate	0.094	0.185	0.239	0.465	0.202	0.042



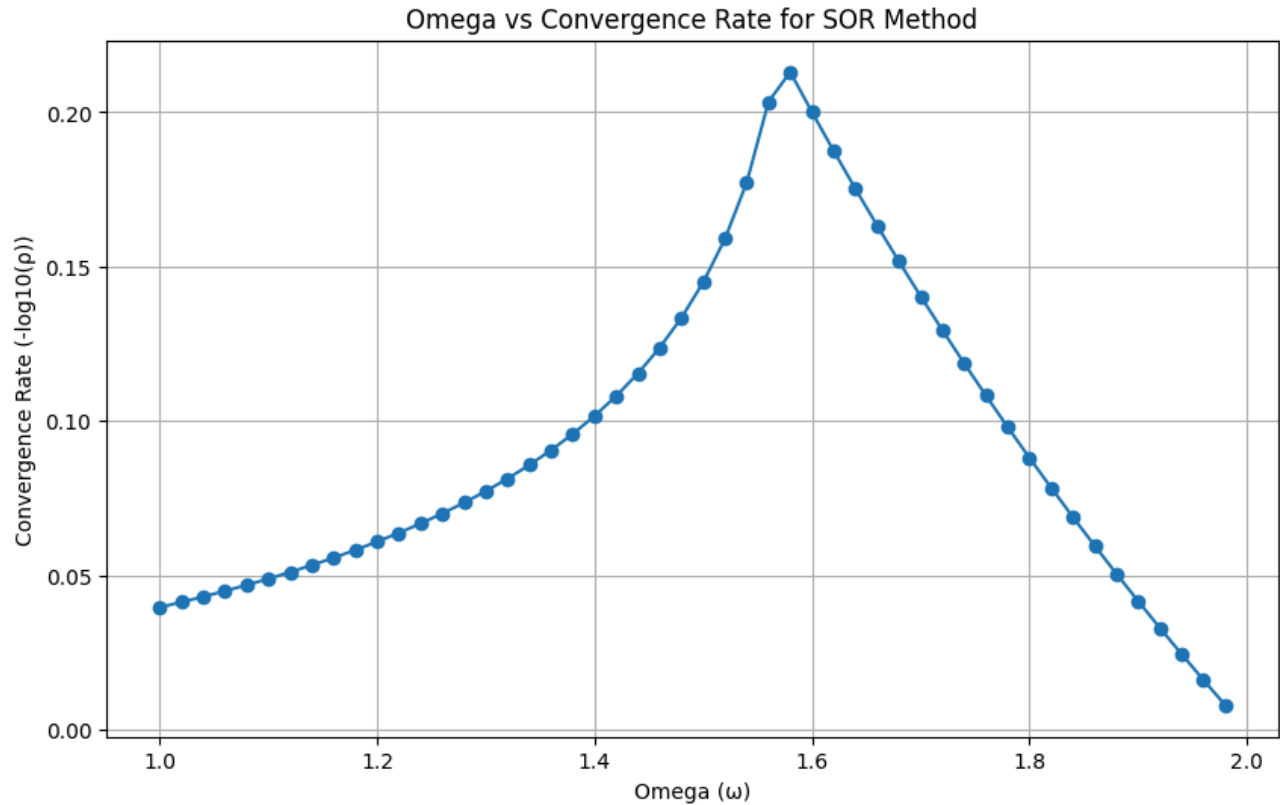
77 points on the mesh:

	Jacobi	Gauss-Seidel	GS-SOR			
			$\omega=1.1$	$\omega=1.3$	$\omega=1.6$	$\omega=1.9$
Iterations till 1.0E-06	97	52	42	25	34	171
Spectral Radius	0.883	0.781	0.734	0.583	0.631	0.909
Convergence Rate	0.054	0.107	0.134	0.234	0.199	0.041



163 points on the mesh:

	Jacobi	Gauss-Seidel	GS-SOR			
			$\omega=1.1$	$\omega=1.3$	$\omega=1.6$	$\omega=1.9$
Iterations till 1.0E-06	248	132	109	71	35	181
Spectral Radius	0.955	0.912	0.893	0.837	0.631	0.908
Convergence Rate	0.020	0.039	0.049	0.077	0.200	0.041



Conclusion and Discussion:

1. The Gauss-Seidel method converges faster than the Jacobi iteration by almost two times for all mesh configurations.
2. Successive Over Relaxation(SOR) method achieves faster convergence (than Gauss-Seidel and Jacobi) for a range of values of the relaxation factor ω . Choosing a very low or too high value of ω degrades the performance.
3. As expected the number of iterations needed to solve the equations increases with the number of mesh points.
4. For 45, 77 and 163 points, the SOR method gave very good results for values of 1.3, 1.3 and 1.6. From the plots we observe that the optimal relaxation factors for meshes with 45, 77 and 163 points are around 1.27, 1.4 and 1.58 respectively hence verifying the previous observations.
5. Although in almost all the cases, lower number of iterations meant a higher convergence rate, an anomaly was observed for the mesh with 163 points. Although the Gauss-Seidel method converged for a lower number of iterations, its convergence rate was lower than that of SOR method with $\omega = 1.9$.